

difference of 1065kHz and 1520kHz as 455kHz, which is same as 455kHz produced in earlier case!

This means two stations will be extracted at the same time on 455kHz, one from 610kHz and another from 1520kHz.

In this case 1520kHz signal is known as ‘image of the unwanted signal frequency,’ and many times it is simply referred to as ‘image frequency.’

This image frequency needs to be prevented from entering the mixer.

One way to do that is having a tuning circuit to select the incoming RF signals. The tuning circuit may use a variable capacitor (Fig. 5) to select a desired frequency. However, one must manually change the variable capacitor responsible for frequency selection at the RF front-end according to the frequency of the LO. That is cumbersome.

Therefore, we need a mechanism where the capacitor of the tuning circuit should automatically be modified, thereby allowing a particular RF signal of our

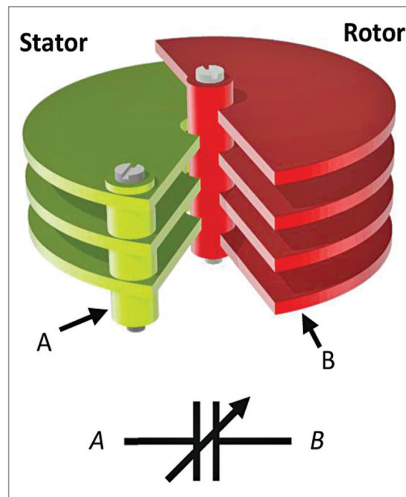


Fig. 5: A variable capacitor

desired frequency in relative of the frequency produced by the LO. Therefore, many systems have twin gang variable tuning capacitors, which consist of two separate variable capacitors whose value can be varied together (not independently) mechanically (see Fig. 6). Thus, one capacitor is used in RF tank circuit on the RF front-end, and another is used to control the frequency of the local oscillator (LO). When changed, capacitance of both the capacitors is changed simultaneously.

Thus, when the RF front-end circuit is tuned for selecting the 610kHz with the help of one variable tuning capacitor, for example, the local oscillator (LO) will automatically (being mechanically coupled) produce 1065kHz with the help of the second variable tuning capacitor. Since the RF front-end is tuned to pick 610kHz, it is not picking other frequencies, including the unwanted image frequency 1520kHz!

Thus the image frequency has no chance to enter the mixer.

Future designs

Modern microprocessors allow building of a software-defined radio (SDR), also known as software radio

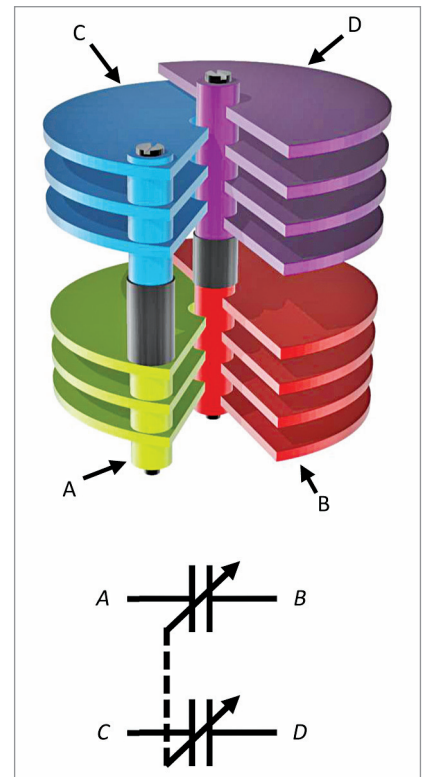


Fig. 6: Two separate variable capacitors mechanically connected together to vary together

architecture, where the IF processing after the initial IF filter is implemented in software. (See <https://www.electronicsforu.com/electronics-projects/software-defined-radio-with-android-smartphones>.)

Many low-cost FM radios have now incorporated the SDR architecture. With greater advancements in the capabilities of the analogue-to-digital converters and the digital signal processing (DSP) capabilities of microprocessors or the dedicated DSP processors, the incoming modulated carrier signal is often directly sampled instead of first getting converted into the IF and then sampled.

Regardless of the advancements, the journey and the concept of IF and the image frequency are fascinating. They are fundamental and relevant even in our modern telecommunications, both in audio and video. **EFY**

Image frequency calculation

It is well accepted that:

Image frequency = Receiving RF frequency + (2 x intermediate frequency)

But how? Let us say we have received an RF broadcasting of an AM signal at frequency R1 with local oscillator frequency L. That means:

$$IF = L - R1$$

$$\text{Therefore, } L = R1 + IF$$

The image frequency is the unwanted image of the signal transmitted at frequency R2. When this R2 is mixed in the mixer with the oscillator frequency L, this should also produce IF.

$$IF = R2 - L$$

$$\text{Therefore, } L = R2 - IF$$

$$\text{Thus, } R1 + IF = R2 - IF$$

Solving for R2 (image frequency), we get,
 $R2 = R1 + 2(IF)$